

Linear Regression

HW 2

Due 9/8 at 11:59pm

Directions: Submit a .pdf file containing your responses for the homework. The .pdf can be converted from Latex file, pictures of your handwriting solutions, word files, markdown files, etc (anything that can be converted into .pdf). If there are coding problems, only include the theory responses question in the .pdf, and upload a separate notebook for Python code.

Written Questions:

1. A student working on a summer internship in the economic research department of a large corporation studied the relation between sales of a product (Y , in million dollars) and population (X , in million persons) in the firm's 50 marketing districts. The classical regression model with normal errors was employed. The student first wished to test whether or not a linear association between Y and X existed. The student accessed a simple linear regression program and obtained the following information on the regression coefficients:

Parameter	Estimated Value	95 Percent Confidence Limits	
Intercept	7.43119	-1.18518	16.0476
Slope	0.755048	0.452886	1.05721

- a. The student concluded from these results that there is a linear association between Y and X . Is the conclusion warranted? What is the implied level of significance? **Sol:** Yes. Since it does not contain zero, the 95% CI for β_1 is [0.452886, 1.05721]. This shows evidence of a significant association between Y and X with the probability of a Type 1 error $\alpha = 0.05$.
 - b. Someone questioned the negative lower confidence limit for the intercept, pointing out that dollar sales cannot be negative even if the population in a district is zero. Discuss. **Sol:** We are 95% confident that the true value of β_0 falls between [-1.18518, 16.0476]. This does not mean it equals -1.18518.
2. A member of a student team playing an interactive marketing game received the following computer output when studying the relation between advertising expenditures (X) and sales (Y) for one of the team's products:

Estimated regression equation: $\hat{Y} = 350.7 - .18X$
Two-sided P -value for estimated slope: .91

The student stated: "The message I get here is that the more we spend on advertising this product, the fewer units we sell!" Comment.

Sol: False. Since $p = 0.91 > \alpha = 0.05$, we cannot conclude that X is a significant predictor of Y . Thus, we cannot comment with evidence on how X would impact Y .

3. For each of the following questions, explain whether a confidence interval for a mean response or a prediction interval for a new observation is appropriate.
 - a. What will be the humidity level in this greenhouse tomorrow when we set the temperature level at 31°C? **Sol:** Prediction interval

- b. How much do families whose disposable income is \$23,500 spend, on the average, for meals away from home? **Sol:** Confidence interval
- c. How many kilowatt-hours of electricity will be consumed next month by commercial and industrial users in the Twin Cities service area, given that the index of business activity for the area remains at its present level? **Sol:** Prediction interval
4. Using the classical regression model with normal errors in an engineering safety experiment, a researcher found for the first 10 cases that R^2 was zero. Is it possible that for the complete set of 30 cases R^2 will not be zero? Could R^2 not be zero for the first 10 cases, yet equal zero for all 30 cases? Explain.

Sol: (i) If R_{10}^2 is zero, it's still possible that R_{30}^2 is not zero.

$$R_{10}^2 = \frac{\sum_{i=1}^{10} (\hat{y}_i - \bar{y}_{10})^2}{SST_{10}} = 0 \Leftrightarrow \sum_{i=1}^{10} (\hat{y}_i - \bar{y}_{10})^2 = 0 \text{ each } \hat{y}_i = \bar{y}_{10} \text{ for } i = 1, 2, \dots, 10.$$

$R_{30}^2 = \frac{\sum_{i=1}^{30} (\hat{y}_i - \bar{y}_{30})^2}{SST_{30}}$ is using different terms and a different $\bar{y}$, so it can still result in a non-zero R^2 .

(ii) If R_{10}^2 is not zero, $R_{30}^2 = 0$ is still possible.

5. Recall the following statement about in-sample predictions: assuming (x_i, y_i) belongs to the training sample we have used to fit the SLR, then the residual e_i has mean 0 and variance $\sigma^2[1 - \frac{1}{n} - \frac{(x_i - \bar{x})^2}{SSX}]$. Now imagine after fitting a regression line and obtaining the parameter estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, we are predicting an observation that's out of sample, i.e. the observation has independent variable value $X = x_0$, where x_0 does not belong to the data used to fit the model. The true response variable

$$y_0 = \beta_0 + \beta_1 x_0 + \epsilon_0$$

is predicted by

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

What is the mean and variance of the out-of-sample prediction bias $e_0 = y_0 - \hat{y}_0$? Hint: since this is an out-of-sample point, ϵ_0 is uncorrelated with all the in sample random errors ϵ_i , $i = 1, \dots, n$.

Sol: Given a new $X = x_0$

$$\begin{aligned} e_0 &= y_0 - \hat{y}_0 = \beta_0 + \beta_1 x_0 + \epsilon_0 - \hat{\beta}_0 - \hat{\beta}_1 x_0 \\ E(e_0) &= \beta_0 + \beta_1 x_0 + E(\epsilon_0) - E(\hat{\beta}_0) - E(\hat{\beta}_1) x_0 \\ &= \beta_0 + \beta_1 x_0 + 0 - \beta_0 - \beta_1 x_0 = 0. \end{aligned}$$

Then,

$$\begin{aligned} Var(e_0) &= Var(\beta_0 + \beta_1 x_0 + \epsilon_0 - (\bar{y} - \bar{x}\hat{\beta}_1) - \hat{\beta}_1 x_0) \\ &= Var[\beta_0 + \beta_1 x_0 + \epsilon_0 - \bar{y} - (x_0 - \bar{x})\hat{\beta}_1] \\ &= Var(\epsilon_0) + Var(\bar{y}) + (x_0 - \bar{x})^2 Var(\hat{\beta}_1) - 2Cov(\epsilon_0, \bar{y}) - 2(x_0 - \bar{x})Cov(\epsilon_0, \hat{\beta}_1) + 2(x_0 - \bar{x})Cov(\bar{y}, \hat{\beta}_1) \end{aligned}$$

Since ϵ_0 is uncorrelated with all $\epsilon_1, \dots, \epsilon_n$. ϵ_0 is uncorrelated with $y_1, \dots, y_n$. So $Cov(\epsilon_0, \hat{\beta}_1) = 0$.

We also know that

$$\begin{aligned} Cov(\bar{y}, \hat{\beta}_1) &= Cov\left(\frac{1}{n} \sum_{i=1}^n y_i, \sum_{j=1}^n k_j y_j\right) \\ &= \sum_{j=1}^n \frac{k_j}{n} Cov(y_j, y_j) \\ &= \frac{\sigma^2}{n} \sum_{j=1}^n k_j \\ &= 0. \end{aligned}$$

$$\begin{aligned}
&\text{Knowing this, } Var(e_0) = Var(\epsilon_0) + Var(\bar{y}) + (x_0 - \bar{x})^2 Var(\hat{\beta}_1) \\
&= \sigma^2 + \frac{\sigma^2}{n} + (x_0 - \bar{x})^2 \frac{\sigma^2}{S_{xx}} \\
&= \sigma^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]
\end{aligned}$$

Coding questions:

1. The data "hospital_infection.csv" recorded the patients infection rates data from 58 hospitals. Let's take **InfctRisk** as the response and **Stay** (average staying days) as the predictor and perform a regression analysis using Python. You are not restricted to the libraries and functions I have used in class. Upload your python file in the homework submission. In your .pdf, answer the following questions:

(a) From scatter plot: is there any initial relationship you can observe? i.e. linear or not, positive or negative, etc.

Sol: a. There does seem to be a weak to moderate positive linear relationship from the scatterplot.

(b) From simple linear regression: write down the equation of the fitted line.

Sol: b. $y = -1.1598 + 0.5689x$ OR **InfctRisk** = $-1.1598 + 0.5689 * \text{Stay}$

(c) From the summary: what is the test statistic and p-value of the coefficient of **Stay**. How does that suggest the impact of **Stay** to **InfctRisk**?

Sol: c. $t_{stat} = 6.041$, p-value < 0.0001 . This suggests Stay has a significant impact on InfectionRisk.

(d) Calculate the 95% confidence interval for β_1 . Does it match with the result from the summary output?

Sol: d. $\hat{\beta}_1 \pm t_{\frac{\alpha}{2}, n-2} S(\hat{\beta}_1) = 0.5689 \pm 2.003 \sqrt{\frac{MSE}{S_{xx}}} = [0.3803, 0.757]$

(e) Run ANOVA: what is the test statistic and p-value of the F test? What does that suggest about the significance of your model?

Sol: e. $F_{stat} = 36.49$, p-value = 1.3×10^{-7} . This suggests the model is significant compared to a null model without x.

(f) Verify that squaring the t-statistic associated with testing the null hypothesis $H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 \neq 0$ is the same value as F-statistic generated from the ANOVA. Also verify that the p-values are approximately the same for the two tests.

Sol: f. $t_{stat}^2 = 36.49 = F_{stat}$

(g) From the summary: what is the value of R^2 ? What does it suggest about the goodness-of-fit of your model?

Sol: g. $R^2 = 0.395$. This suggests that 39.5% of the variance in Y is explained by the model fitted. It's a relatively low measurement for the "goodness of fit", but to have an objective comparison further analysis is needed on this data.

(h) Predict the infection risk for a hospital with an average stay length of 32 days.

Sol: h. $\hat{Y}|_{\text{Stay}=32} = 17.044$

2. Continue with the hospital infection example from above. Use python to draw the residual v.s. fitted value plot and QQ plot based on your model. Based on your plots, evaluate the following assumptions. A brief/roughly observed comment is acceptable (i.e., seems to be false because...; hard to judge because...; etc.).

(a) Infection risk and stay length have a linear relationship.

Sol: There is no obvious change of pattern (trend going up and down), so we can assume a linear relationship is valid.

(b) The assumption of error terms have constant variance is valid.

Sol: It seems to have a change bandwidth at the end of the plot, not significant in other parts. An assumption of constant variance might be okay but need further examination.

(c) The assumption of error terms are independent is valid.

Sol: For the assumption of independence to be significantly violated we should observe an obvious upward or downward trend. Here it seems valid.

(d) The assumption of error terms are normally distributed is valid.

Sol: The assumption of normality is not significantly violated. Further testing might be needed.

3. Continue with the hospital infection example from above. Use Python to give the 95% confidence interval for the mean infection risk when the average stay is 32 days, and the 95% prediction interval of the infection risk for a new patient who would stay 32 days.

Sol: